Project Evaluation Results

Datamites Project Feedback Report

Team	PTID-AIE-JUL-26-11142	**Project**	PRAICP-1004
Course	AIE	**Overall Grade**	A
Topic	Rainfall Time Series		

Dear Team Member,

Please find below the detailed evaluation feedback for your project submission. Review each section carefully and use the improvement suggestions to guide your next steps.

Data Understanding

Present Implementation

The submission provides a thorough overview of the dataset, including visualization of trends, seasonality (via `seasonal_decompose`), and correlation between variables. Data types, missing values, and stationarity (ADF test) are explicitly checked and documented. The connection between the domain (meteorology) and the statistical properties of the data is well-articulated.

Scope for Improvement

While the seasonal decomposition is helpful, the submission could further analyze the residuals to confirm if the models are capturing all deterministic patterns or if white noise remains, which would provide better insight into the predictability limits of the series.

Data Preprocessing

Present Implementation

The submission correctly identifies the time-series nature of the data and avoids standard k-fold cross-validation in favor of chronological splitting and TimeSeriesSplit. Feature engineering is logical, utilizing lags and rolling windows to transform the problem into a supervised learning task. The handling of indices and frequency (MS) is appropriate.

Scope for Improvement

The submission could explicitly demonstrate the impact of differencing or log-transformations on the stationarity of the series prior to feeding it into the ML models, as some regressors may struggle with non-stationary inputs.

Model Development

Present Implementation

The project demonstrates excellent breadth by comparing five distinct models: a statistical model (SARIMA) and four machine learning models (Linear Regression, Random Forest, Gradient Boosting, XGBoost). The use of hyperparameter tuning with an appropriate time-series-aware cross-validation strategy is commendable. Performance metrics (MAE, RMSE, R2) are applied consistently across all models.

Scope for Improvement

The submission uses different approaches for the 1-year forecast (SARIMA vs. Recursive ML). Providing a more detailed explanation of why the recursive strategy for ML might be more prone to error accumulation (compounding of forecast errors) would add depth to the model selection justification.

Results & Conclusion

Present Implementation

Results are clearly presented using tables and visualizations. The business implications regarding rainfall management are directly linked to the technical findings. Limitations, such as the inherent noise in meteorological data and the constraints of monthly-aggregated data, are acknowledged honestly.

Scope for Improvement

Include a brief discussion on feature importance from the Random Forest or XGBoost models to provide further transparency into which specific lags or calendar features were the primary drivers of the predictions.

Code Quality & Reproducibility

Present Implementation

The code is modular, well-commented, and follows a logical structure. The use of clear section headers makes the notebook easy to navigate. Variable naming is intuitive, and the workflow is fully reproducible given the provided dataset source.

Scope for Improvement

No major improvements needed in code structure. Adding a requirement.txt file or listing library versions would enhance the long-term reproducibility of the environment.

Overall Feedback

Present Implementation

This is a highly professional and rigorous submission. The analysis demonstrates a strong grasp of both classical time-series analysis and machine learning workflows. The decision to incorporate SARIMA alongside modern ensemble methods is a significant strength, as it allows for an objective comparison of different forecasting philosophies. The documentation of challenges faced and the thoughtful consideration of time-series constraints (e.g., avoiding data leakage) show a sophisticated understanding of the subject matter.

Scope for Improvement

The project is already very strong. Future work could explore incorporating exogenous variables (like climate indices mentioned in the conclusion) to see if model accuracy can be further improved.

Best regards,
LMT